Colombia Coffee- Prototype Indexes
IRI, Earth Institute at Columbia University

Overview

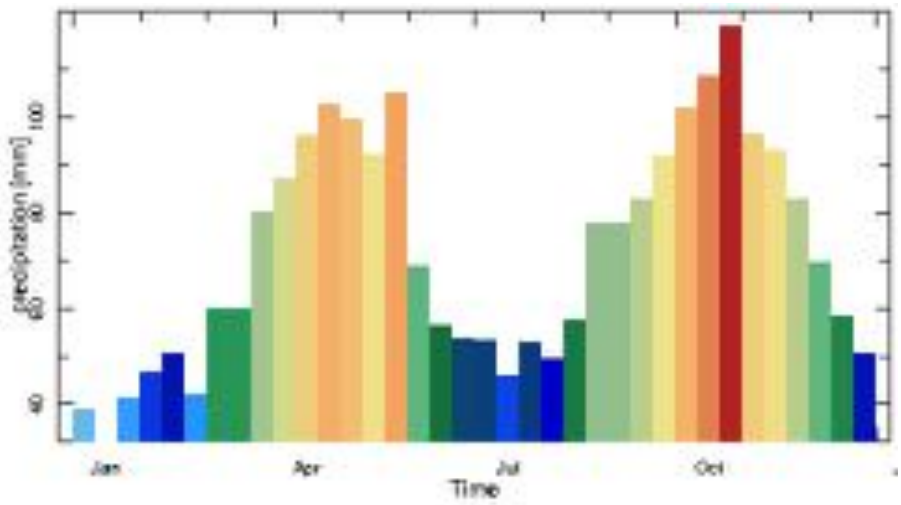
This document outlines the process carried out to develop the prototype indexes for two coffee regions in Colombia—Caldas and Cauca. During discussions with several Nespresso farmer groups in both regions, drought and excess rainfall were identified as the main potentially indexable climate risks for coffee production. According to the farmers, the timing during which coffee production is the most vulnerable to drought is preceding the harvest period, because the coffee bean needs enough water before harvest to mature enough to be picked. Coffee production is also vulnerable to excess rainfall, more so during the key flowering periods. For more details about specific risks, refer to Risk Assessment document submitted May 2015.

The following sections outline the index structure and process for the drought prototype product. It has been identified that through additional analysis and discussions with the local experts and farmers the index's accuracy at targeting identified risks could improve. Additionally, further analysis will provide better insight about the possibility of a prototype index for excess rainfall. The following analysis and process was carried out using UCSB CHIRPS rainfall estimates, as after careful consideration of available datasets it is the most representative of conditions on the ground (including available weather stations) and it will be operational into the future (as opposed to other satellite technology available in the region, such as TRMM.)

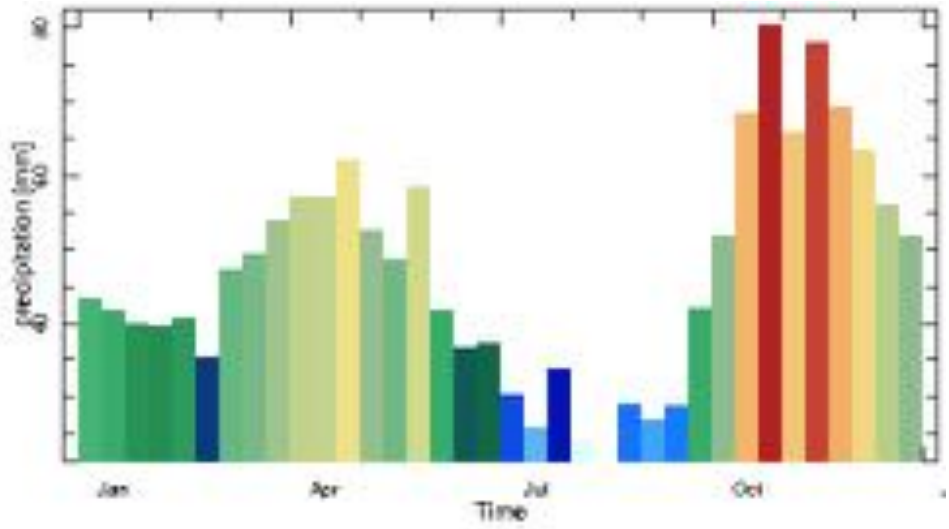
Index Structure and Process

1) Climatology and Index Windows
    a. Caldas
       For the Caldas region, there are two main periods during which coffee is harvested: Between February and April and between August and October. Below find a representation of the region's climatology, using CHIRPS data on the IRI Data Library.

       Therefore, the identified drought peril during harvest requires a two-window index covering the months of January through March and July and August, corresponding to dekads 4 through 9 and 19 through 24.



b. Cauca
In Cauca, there is one harvesting period: between March and May. The windows of the index will therefore correspond to dekads 7 through 15. Below is a representation of the climatology for the region using CHIRPS data on the IRI Data Library.



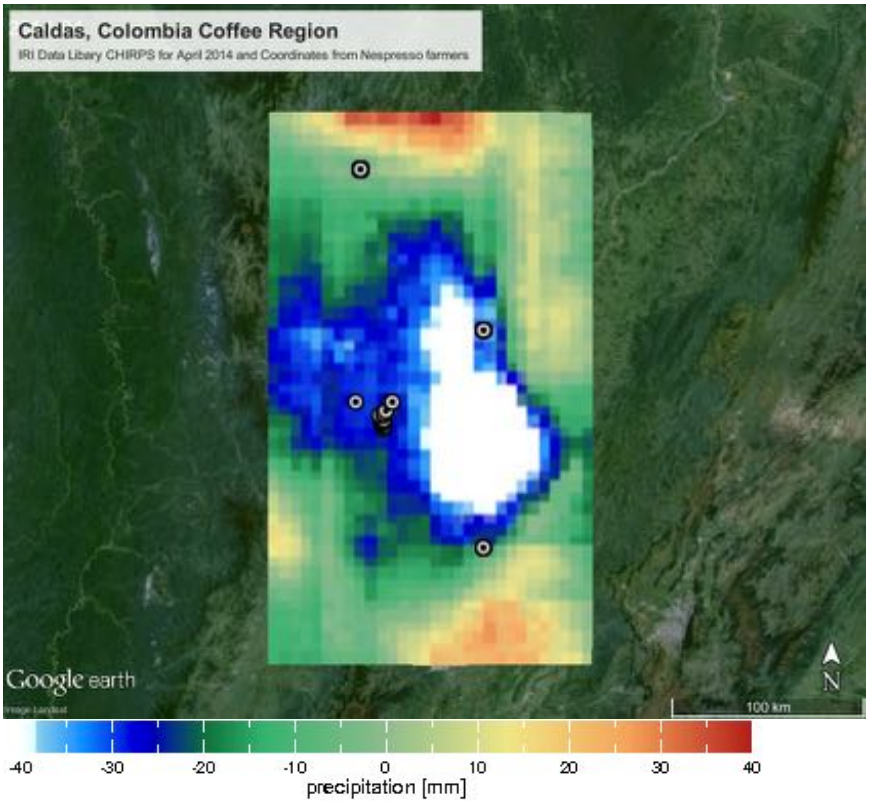
2) Regional boxes
The coffee producing regions in Colombia have homogenous altitudes. Through discussions with farmers it was identified that at different altitudes coffee production experiences climate risks at varying degrees. For example, at higher altitudes excess rainfall tends to be a more severe risk, generally leading to greater losses than at lower altitudes; while at lower altitudes droughts are a greater risk to production (since low regions tend to be generally drier.) For this reason, altitude variations is as a key element to take into account for developing the index, as we would expect rainfall patterns to have varying impacts.

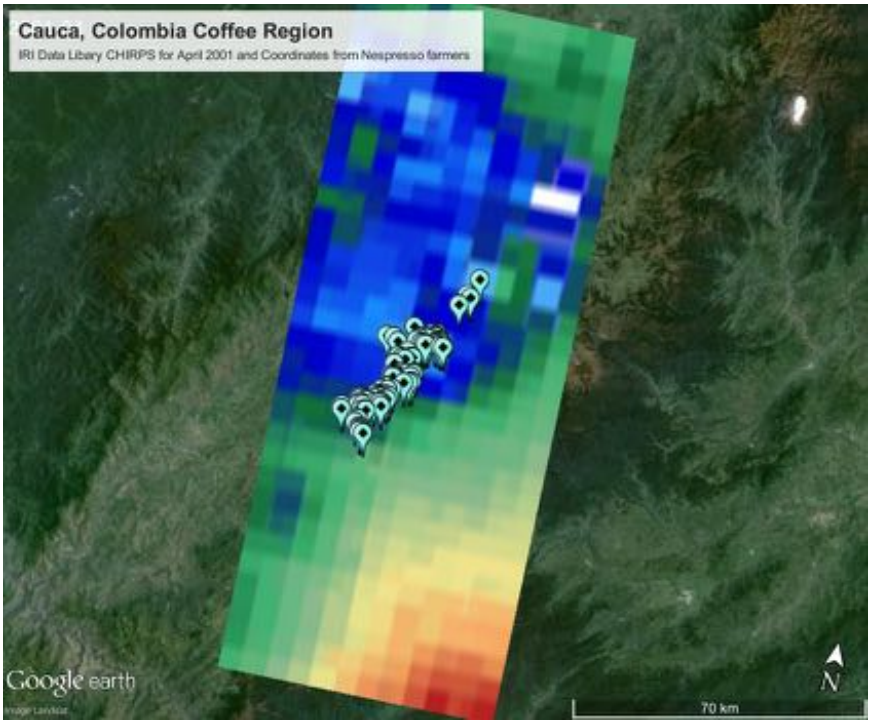
Below we outline the process carried out to select regions according altitude levels, which generally correspond to differing rainfall patterns, in both Caldas and Cauca regions.



1) Identifying patterns in climate conditions (some areas have stronger impacts than others of either drought or excess rainfall.) For example, see representation of rainfall anomalies in the Caldas region for the month of April 2014, which was identified as a dry period by the farmers.

We also include a similar representation for the Cauca region, which also shows homogenous precipitation patterns. Below is a visual representation of April 2001, which is one of the years reported as bad by the farmers.





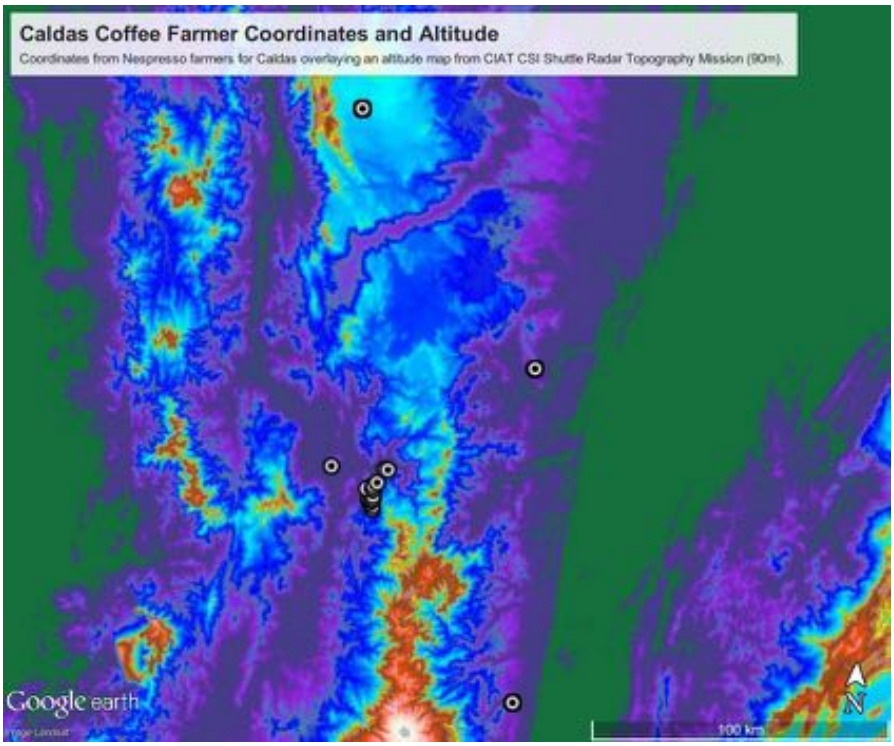
For additional representations of the rainfall anomalies follow this link to the IRI Data Library CHIRPS viewer for Caldas, and the following link for Cauca.

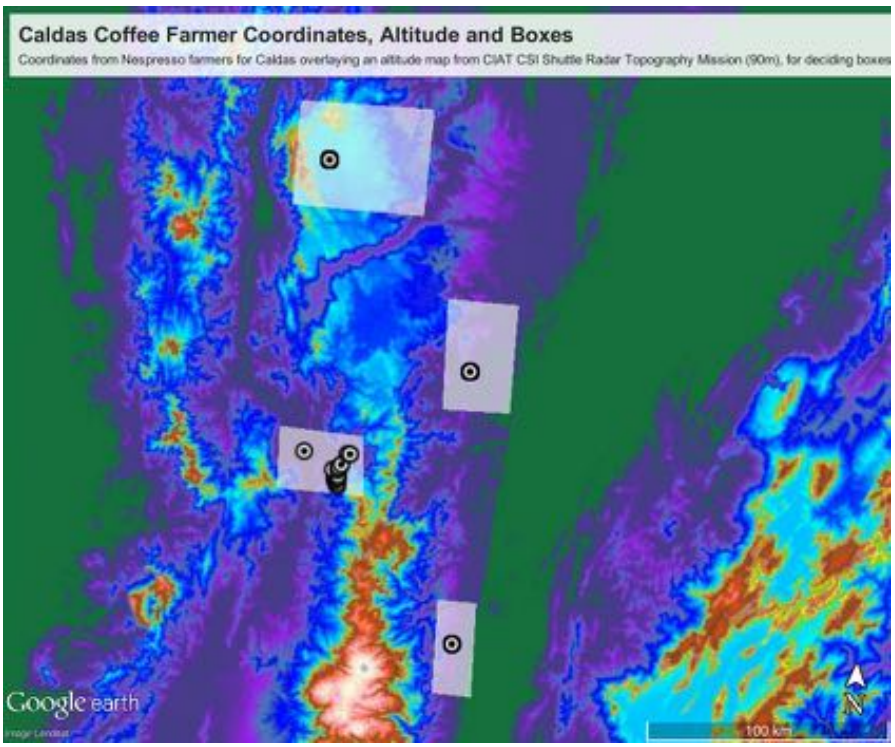
2) Referencing altitude maps to identify homogeneity among coordinate points with respect to altitude. Regions were selected using the CIAT SRTM altitude dataset in the IRI Data Library. Below find the identification process for each region, using Nespresso farmer's coordinates overlaying the altitude levels.

   Using a qualitative approach to analyze the differences within the regions, we identified four regions within Caldas and six smaller regions within Cauca. These boxes were determined based on both climate patters and altitude levels within the regions, expecting similar risks faced within the boxes.

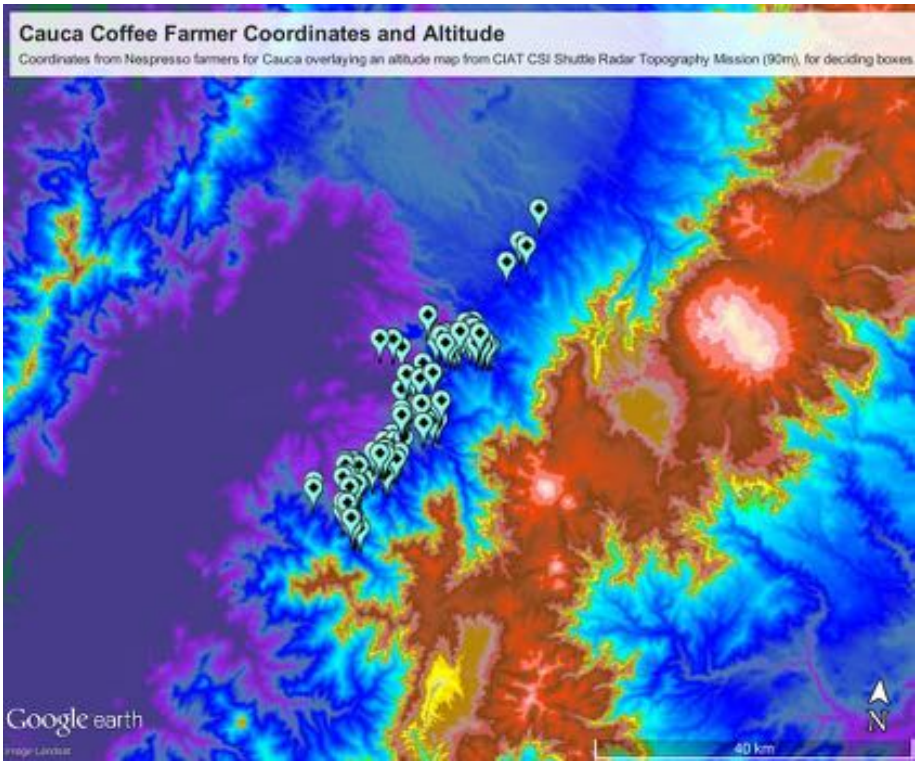
   Additionally, in the Annexed tables please find the coordinates for the 10 boxes used to develop the prototype product..

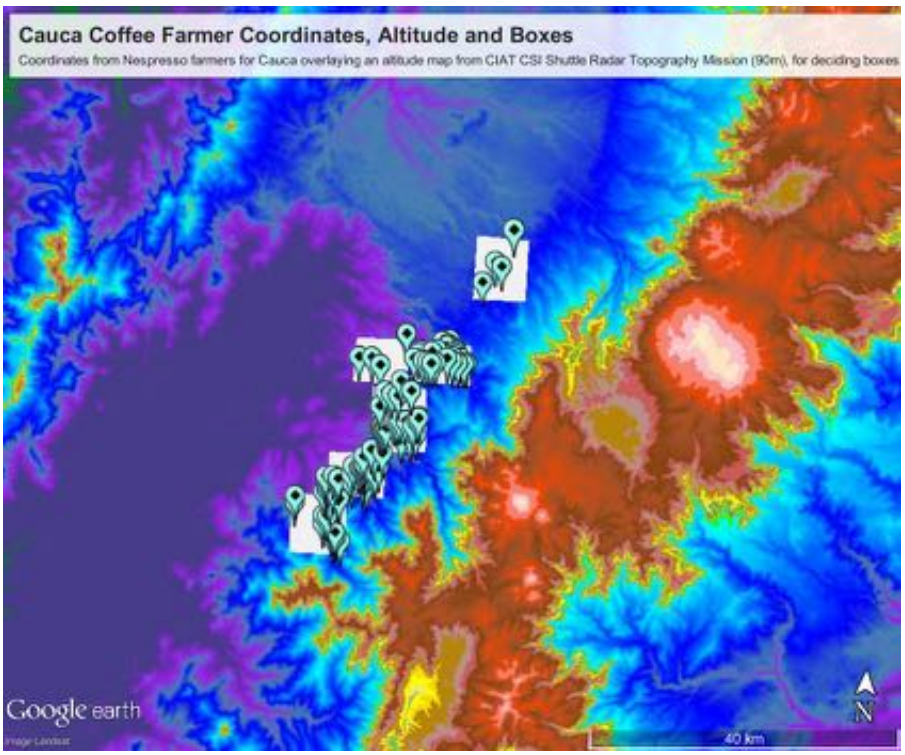
   Caldas Coordinates and CIAT SRTM Altitude Map



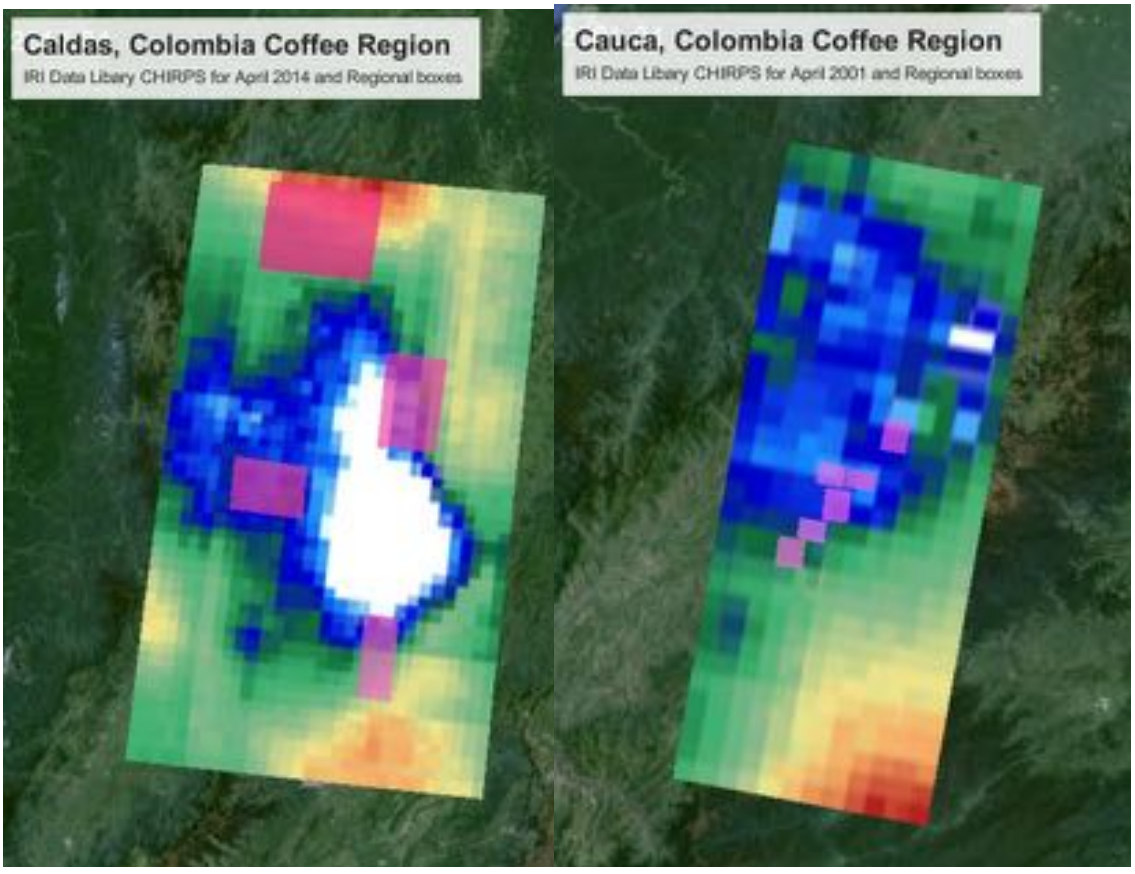


Cauca Coordinates and CIAT SRTM Altitude Map





Overview of boxes and climate patterns due to altitude levels



3) Index Structure and Calibration:
The index is triggered using the USCB CHIRPS version 2.0 satellite estimates of rainfall, freely available from IRI or UCSB (links to IRI Data Library above). For each, the satellite information **was spatially averaged dataset at a dekadal temporal resolution.** Datasets are available upon request.

The start and end of each index is set by considering the following:

1. Local experts opinion through discussions with the farmer groups
2. The required precision of satellite measurements during the window;
3. The rainfall patterns that persist through the window during a normal season (i.e. the climatology during the window);
4. How the absence of rainfall during the window correlates with actual on-the-ground loss, as reported by farmers;

The trigger is calibrated with the goal of providing a meaningful payout for the worst year out of six in the historical dataset. In trigger and exit calculations, missing data is filled with historical averages for that day. Unless specified otherwise, any supplementary precipitation data provided with this report uses this missing data protocol.

4) Index Parameters

The parameters required for the index are: start dekad, end dekad, cap, trigger, and exit. Each index consists of a single window of consecutive dekads. The calendar dekads used are presented in the Calendar Dekad Translation tables at the end of this document.

To calculate payouts, the satellite rainfall estimate for each dekad in the index window is totaled. If any dekad has a rainfall total above the cap, that dekad's rainfall is set to the index cap. The capped rainfall of all of the dekads in the index window are then added together to arrive at the Capped Rainfall Sum.

The formula below can be used to determine the payout for any maximum liability: Payout= (1 – ((Capped Rainfall Sum – Exit) / (Trigger – Exit)))*Max Liability.

APPENDIX

1) Site coordinates

The latitude and longitude coordinates used to reference the boxes are listed in the table below. These are the coordinates utilized to query the satellite databases from the IRI Data Library.

	Name	X1	X2	Y1	Y2
1	Caldas1	-75.692639	-75.205275	6.965319	6.539656
2	Caldas2	-75.095642	-74.843319	6.246192	5.850128
3	Caldas3	-75.688358	-75.365856	5.746278	5.522914
4	Caldas4	-75.068725	-74.923878	5.153489	4.787064
5	Cauca1	-76.659781	-76.600653	2.408003	2.335456
6	Cauca2	-76.791603	-76.730972	2.280669	2.231389
7	Cauca3	-76.724375	-76.662533	2.235556	2.279686
8	Cauca4	-76.766508	-76.704947	2.22625	2.154672
9	Cauca5	-76.75065	-76.813264	2.1479	2.102114
10	Cauca6	-76.855417	-76.789758	2.095264	2.031106

2) Draft Prototype Drought Index Parameters

	Name	Region	Trig	Exit	First	Last
1	Caldas1	Caldas	155	120	1-Feb (4)	31-Mar (9)
2	Caldas1	Caldas	328	251	1-Jul (19)	31-Aug (24)
3	Caldas2	Caldas	281	217	1-Feb (4)	31-Mar (9)
4	Caldas2	Caldas	323	232	1-Jul (19)	31-Aug (24)
5	Caldas3	Caldas	217	189	1-Feb (4)	31-Mar (9)
6	Caldas3	Caldas	218	95	1-Jul (19)	31-Aug (24)
7	Caldas4	Caldas	253	213	1-Feb (4)	31-Mar (9)
8	Caldas4	Caldas	150	73	1-Jul (19)	31-Aug (24)
9	Cauca1	Cauca	440	364	1-Mar (7)	31-May (15)
11	Cauca2	Cauca	395	332	1-Mar (7)	31-May (15)
13	Cauca3	Cauca	395	329	1-Mar (7)	31-May (15)
15	Cauca4	Cauca	379	342	1-Mar (7)	31-May (15)
17	Cauca5	Cauca	382	302	1-Mar (7)	31-May (15)
19	Cauca6	Cauca	382	286	1-Mar (7)	31-May (15)

3) Payout years and comparison with farmer bad years

Caldas

	Farmer Groups Bad Years		PAYOUTS							
			Caldas1	Caldas1	Caldas2	Caldas2	Caldas3	Caldas3	Caldas4	Caldas4
	Travie sa	Cosecha	Early	Late	Early	Late	Early	Late	Early	Late
			0	0	0	0	0	0	0	0
			0	0	0	0	0	82	0	93
			17	0	77	0	7	0	27	0
			0	0	0	0	0	0	0	0
1985	2	2	0	0	0	0	62	0	100	0
1986	0	0	0	23	0	78	0	36	0	13
1987	1	0	54	0	0	0	84	0	0	0
1988	0	0	100	0	0	0	32	0	84	0
1989	0	0	0	0	0	0	0	0	0	0
1990	1	1	0	0	0	0	0	0	0	0
1991	1	0	0	53	0	58	0	20	0	30
1992	1	0	65	0	0	0	100	0	6	0
1993	0	1	0	0	0	0	0	0	0	0
1994	0	0	0	0	0	0	0	0	0	0
1995	0	1	68	0	63	0	35	0	60	0
1996	0	1	0	0	0	0	0	0	0	0
1997	2	1	0	100	0	100	0	100	0	100
1998	1	**6**	0	0	1	0	0	0	0	0
1999	0	0	0	0	0	0	0	0	0	0
2000	0	0	0	0	0	0	0	0	0	0
2001	1	1	0	8	4	44	0	0	0	0
2002	0	2	0	1	0	6	0	4	0	1
2003	0	1	0	0	0	0	0	0	0	0
2004	0	0	0	0	0	0	0	0	0	0
2005	0	2	65	0	1	0	0	0	0	0
2006	0	1	0	0	0	0	0	0	0	0
2007	1	1	0	0	100	0	0	0	45	0
2008	**2**	**3**	0	0	0	0	0	0	0	0
2009	1	2	0	0	0	0	0	0	0	0
2010	1	**7**	0	0	0	0	0	0	0	0
2011	**3**	**3**	0	0	0	0	0	0	0	0
2012	2	1	0	0	0	0	0	0	0	0
2013	2	1	0	0	0	0	0	0	0	0
2014	**4**	2	0	32	0	34	0	44	0	69

Cauca

	Farmer Bad Years Cosecha (Nov-Dec)	Cauca1 Pay	Cauca2 Pay	Cauca3 Pay	Cauca4 Pay	Cauca5 Pay	Cauca6 Pay
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
		0	0	0	0	0	0
1985	2	58	99	61	73	58	61
1986	0	0	0	0	0	0	0
1987	0	0	0	0	0	0	0
1988	0	0	0	0	0	12	26
1989	0	0	0	0	0	0	0
1990	1	0	0	0	0	0	0
1991	0	0	0	0	0	0	0
1992	2	16	0	0	0	0	0
1993	0	0	0	0	0	0	0
1994	1	0	0	0	0	0	0
1995	1	0	0	0	0	0	0
1996	1	0	0	0	0	0	0
1997	0	1	11	11	2	0	0
1998	3	0	0	0	0	0	0
1999	4	2	0	0	0	0	0
2000	2	0	0	0	0	0	0
2001	0	99	24	66	28	24	29
2002	0	0	0	0	0	0	0
2003	0	0	64	49	54	9	2
2004	1	0	0	0	0	0	0
2005	3	0	0	0	0	0	0
2006	0	0	0	0	0	0	0
2007	0	0	0	0	0	0	0
2008	3	0	0	0	0	0	0
2009	5	0	27	31	44	31	34
2010	2	0	0	0	0	0	0
2011	5	0	0	0	0	0	0
2012	0	57	95	99	100	100	100
2013	2	0	0	0	0	0	0
2014	1	0	0	0	0	0	0

- Years where there are no payouts, but farmers mentioned are recurrently years, are mostly those years only mentioned by one group. Therefore there we are confident that there is meaningful agreement between the indexes and the farmer on-the-ground losses.

- In Caldas, considering these two periods for drought vulnerability a strong enough index through which many of the historical bad years mentioned by the farmers would have received a payout in the past. Payouts for all years, except 1988, agree with the years mentioned by farmers as bad for production. Additional years were also mentioned by farmers, however the farmer bad year reports include both drought and excess rainfall perils.
- For Cauca, two out of four of the meaningful payout years are not in agreement with a year mentioned by the farmers. Through follow-up discussions with farmers and local experts, we could be able to identify how to refine the index so that it better targets coffee's drought risk.